

Demo Day & the Road to HPC

Assemble the Studio, present your work, and race matrix multiplication

Day 10 · Week 2

🎲 Demo Day · Full Board

1 Mission briefing

Ten days ago the Studio was empty. Today it is a full board of seven AI capabilities you built with your own hands. Demo Day runs in three acts: **assemble** the whole Studio, **present** it like a scientist, then run the **Grand Benchmark** that reveals the single operation hiding beneath all of it — and points straight at the world of high-performance computing. *Previously:* you unlocked 🎮 Game Master — agents that learn from reward alone.

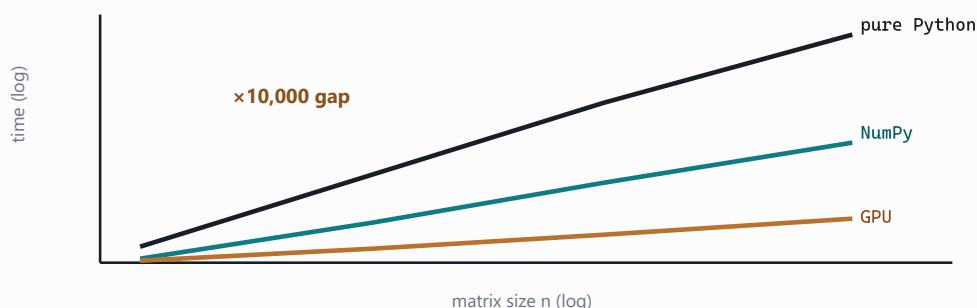
- ☐ Assemble all **7 modules** into one running Studio
- ☐ Present a tight **5-minute demo** with honest model cards
- ☐ Benchmark matmul: **pure Python vs NumPy vs GPU**
- ☐ See **$O(n^3)$** and a **$\times 10,000$** speedup for yourself
- ☐ Connect every module to `C = A @ B`
- ☐ Keep the plot — it's your ticket into **HPC**

2 Field guide the three acts

Act I — Assemble. Every notebook exported its model files into `ai_studio/models/`. Launch the Studio and each satisfied module unlocks its tab. Ten files, seven modules, one app.

Act II — Present. A demo is a science talk: **hook** → two favorite modules **live** → your **hardest bug** → what **surprised** you. A good **model card** states honestly what a model does, how accurate it is, and where it fails. Honesty is the science.

Act III — the Grand Benchmark. Multiply two big matrices three ways — a hand-written triple loop, NumPy, and the GPU — then time them and plot the result on log-log axes. The cost grows like **$O(n^3)$** , and the GPU comes out $\sim 10,000\times$ faster than the loop.



The money plot: the same math, three speeds. On log-log axes each method is a straight line — and the GPU runs about $10,000\times$ faster than the hand-written loop.

📊 **Matrix Watch — the payoff.** Every conv, every attention head, every Q-update you built runs on `C = A @ B`. Matrix multiplication was the whole game all along — and high-performance computing makes it scream. **Keep this plot.**

3 Lab missions your Demo Day checklist

ACT I · ASSEMBLE THE STUDIO

MODULE	FROM	MODEL FILE(S)	✓
 Prediction Machine	Day 2	<code>prediction_machine.joblib</code>	☐
 Pattern Finder	Day 3	<code>pattern_finder.json</code>	☐
 Digit Reader	Day 5	<code>digit_reader.pt</code>	☐
 Photo Detective	Day 6	<code>photo_detective.pt</code> · <code>photo_detective_classes.json</code>	☐
 RPS Arena	Day 7	<code>rps_arena.pt</code> · <code>rps_labels.json</code>	☐
 Language Lab	Day 8	<code>language_lab.json</code>	☐
 Game Master	Day 9	<code>game_master_qtable.npy</code> · <code>game_master_dqn.zip</code>	☐

1 Take the roll

Tick every file in the table above.

Expect: all 10 present in `ai_studio/models/` — nothing missing.

2 How big are your models?

Measure each saved file's size on disk.

Expect: a few KB for the JSON tokens up to tens of MB for the `.pt` models.

3 Launch the Studio

Run `python ai_studio/app.py` from the repo root.

Expect: the banner read **7 / 7 modules unlocked**.

ACT II · PRESENT LIKE A SCIENTIST

1 Write an honest model card

For one module, note what it does, its accuracy, and where it fails.

Expect: real numbers and honest failure cases — not hype.

2 Sketch your 5-minute demo

Hook → two favorite modules live → your hardest bug → what surprised you.

Expect: a run-of-show you can deliver without notes.

ACT III · THE GRAND BENCHMARK

1 Rebuild matmul from memory

Write the triple-loop `matmul` with no peeking.

Expect: it match NumPy exactly on a small test case.

2 Time the hand-written version

Clock your loop, NumPy, and the GPU (best of 3 each).

Expect: milliseconds for NumPy vs whole seconds for the loop.

3 The money plot



Plot time vs n on log-log axes for all three methods.

Expect: three near-straight lines with huge vertical gaps.

4 Put a number on the gap



Divide the slow time by the fast one.

Expect: a speedup on the order of $\times 10,000$.

5 Graduate



Run the final cell and save your money plot — a souvenir worth keeping.

Expect: a saved figure and a send-off to the HPC weeks.

4 Cheat sheet benchmarking matmul

```
for i,j,k: C[i,j] += A[i,k]*B[k,j]
```

Matmul by definition — the $O(n^3)$ triple loop.

```
t = time.perf_counter()
```

Read a high-resolution clock (take best of 3).

```
torch.cuda.synchronize()
```

Wait for the async GPU before you stop the clock.

```
gflops = 2*n**3 / t / 1e9
```

Turn seconds into GFLOPs (a matmul is $2n^3$ ops).

```
C = A @ B
```

The one line every Studio module secretly runs.

```
sys.path.insert(0, str(REPO))
```

Let the notebook import the Studio package.

```
from ai_studio.studio import build_demo
```

Assemble all seven modules in code.

```
build_demo().launch()
```

Bring the whole Studio to life.

5 Light up the full board

The full board — 7 / 7 modules

No new module today. Instead, every tab you built lights up at once. Launch it and confirm all seven are live:

```
$ python ai_studio/app.py
```

✓  Prediction Machine · Day 2

✓  Pattern Finder · Day 3

✓  Digit Reader · Day 5

✓  Photo Detective · Day 6

✓  RPS Arena · Day 7

✓  Language Lab · Day 8

✓  Game Master · Day 9

6 Mini-glossary

benchmark

A controlled speed test comparing methods fairly.

Big-O / $O(n^3)$

How cost grows with size; matmul scales as n cubed.

FLOP / GFLOPs

A floating-point operation · billions of them per second.

vectorization

Replacing Python loops with fast bulk array operations.

HPC

High-performance computing — big math, made fast at scale.

Julia

A fast scientific language built for exactly this kind of number-crunching.

7 Show & tell

1. Which two modules will you demo live, and what's your 10-second hook for each?
2. What was your hardest bug across ten days, and how did you finally kill it?
3. Your triple loop vs the GPU: how many times faster? Why does that gap matter for real AI?

THE LAB

Certificate of Completion

This certifies that

has completed AI Studio — Intro to AI and is hereby recognized
as a

Certified AI Explorer



Prediction Machine



Pattern Finder



Digit Reader



Photo Detective



RPS Arena



Language Lab



Game Master

Instructor

Date

The Lab